we have,

$$P(F_A \cap F_B) = P(F_A \cap F_B | S).P(S) + P(F_A \cap F_B | M).P(M) + P(F_A \cap F_B | \omega).P(\omega)$$

Now,

$$P(F_A \cap F_B | S) = P(F_B \cap F_A | S) = P(F_B | F_A | S) \cdot P(F_A | S)$$

$$= 0.50 \times 0.20$$

Similarly,

$$= 0.1$$

$$P(F_A \cap F_B | M) = 0.15 \times 0.05 = 0.0075$$

$$P(F_A \cap F_B | \omega) = 0.02 \times 0.01 = 0.0002$$

Thus,

$$P(F_A \cap F_B) = 0.1 \times 0.02 + 0.0075 \times 0.2 + 0.0002 \times 0.78$$

$$= 0.003656$$

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(d) If building A failed and B survived, the probability that the earthquake was not strong, i.e. $P[S | F_A \cap \overline{F_B}]$

→ we have,

$$P(F_A \cap \overline{F_B})$$

$$= P(F_A) - P(F_A \cap F_B)$$

$$= 0.0218 - 0.003656$$

$$= 0.018144$$

$$P(S \cap F_A \cap \overline{F_B}) = P(\overline{F_B} \cap F_A \cap S) = P(\overline{F_B} | F_A \cap S) \cdot P(F_A \cap S)$$

$$= P(\overline{F_B} | F_A \cap S) \cdot P(F_A | S) \cdot P(S)$$

we have,

$$P(\bar{E}_1 | E_2) = 1 - P(E_1 | E_2)$$

$$\Rightarrow P(\bar{F}_B | F_A \cap S) = 1 - P(F_B | F_A \cap S)$$

$$= 1 - \frac{P(F_B \cap F_A \cap S)}{P(F_A \cap S)} = 1 - \frac{P(F_B \cap F_A | S)\, P(S)}{P(F_A | S)\, P(S)}$$

$$= 1 - \frac{P(F_B \cap F_A | S)}{P(F_A | S)}$$

$$= 1 - \frac{0.1}{0.2}$$

$$= 0.5$$

$$\therefore P(S \cap F_A \cap \bar{F}_B) = P(\bar{F}_B | F_A \cap S) \cdot P(F_A | S) \cdot P(S)$$

$$= 0.5 \times 0.2 \times 0.02$$

$$= 0.002$$

Now,

$$P(S | F_A \cap \bar{F}_B) = \frac{P(S \cap F_A \cap \bar{F}_B)}{P(F_A \cap \bar{F}_B)} = \frac{0.002}{0.018144}$$

$$= 0.110229$$

$$\therefore \overline{P(S | F_A} \cap \bar{F}_B) = 1 - 0.110229$$

$$= 0.8898$$

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4/ ⓓ → ~~subject of chapter 9.~~

Sol^n.

we have,

$$P''(\theta=\theta_i) = \frac{P(\epsilon|\theta=\theta_i)\,P'(\theta=\theta_i)}{\sum_{i=1}^{k} P(\epsilon|\theta=\theta_i)\,P'(\theta=\theta_i}$$

Where,

$P''(\theta=\theta_i)$ = posterior probability

$P'(\theta=\theta_i)$ = prior probability

Since the capacity of the freeway has been exceeded. The updated PMF of the

ⓐ 300 vph capacity,

$$P''(p=300) = \frac{0.2\times0.25}{0.2\times0.25+0.8\times\left(\frac{(400-350)^2}{40000}\right)}$$

$$= 0.5$$

$$P''(p=350) = \frac{\frac{(400-350)^2}{40000}}{0.2\times0.25+0.8\times\left(\frac{(400-350)^2}{40000}\right)}$$

$$= 0.5$$